

Top 10 AI Model Leaderboard

ETA Shipment
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Safiri AI Take-Home Assignment

Shipment ETA Prediction - Severe Delay Classification - Delay Propagation

Validation ranking; final test excluded from model selection

Purpose. This appendix summarizes the highest-performing AI models for the three assignment tasks. Rankings use the best validation score observed in the completed notebook outputs. ETA uses MAE in minutes, classification uses F1, and propagation records the evaluation protocol beside each score.

BEST ETA MODEL

Extra Trees Regressor

101.736 MAE

BEST SEVERE DELAY MODEL

Extra Trees Classifier

0.9104 F1

BEST PROPAGATION MODEL

Extra Trees Regressor

95.484 MAE

1. ETA Prediction - Top 10 Models

Target: `trip_duration_minutes`. Ranking metric: **Validation MAE**; lower is better. Unit: minutes.

Rank	Model	Model family	Best score
1st	Extra Trees Regressor	Tabular ensemble / trees	101.736 MAE
2nd	Rotation Forest Regressor	Bagging & randomized ensemble	103.838 MAE
3rd	Bagging Regressor	Bagging ensemble	107.432 MAE
4	LightGBM Regressor	Gradient boosting	118.795 MAE
5	Bidirectional LSTM	Sequential neural network	119.246 MAE
6	Random Subspace Regressor	Bagging & randomized ensemble	120.558 MAE
7	Transformer Encoder	Sequential transformer	120.883 MAE
8	Temporal Convolutional Network (TCN)	Sequential convolutional model	120.940 MAE
9	Long Short-Term Memory (LSTM)	Sequential neural network	121.431 MAE
10	Sequence-to-Sequence Transformer	Sequential transformer	122.138 MAE

Interpretation. Extra Trees achieves the lowest validation error and remains close to its final-test MAE, indicating stable generalization under the chronological split.

Report wording. Use: "Extra Trees Regressor achieved the best ETA validation MAE among completed candidate models."

No temporal shuffling

Validation-only selection

Leakage-safe start features

Task definition. Severe delay is defined in the notebook as `delay_minutes > 120`. The 120-minute value is an adjustable SLA proxy rather than an official Delhivery service-level threshold.

2. Severe Delay Classification - Top 10 Models

Target: `severe_delay`. Ranking metric: **Validation F1**; higher is better. Each classifier uses its own validation-optimized probability threshold.

Rank	Model	Model family	Best score
1st	Extra Trees Classifier	Tabular ensemble / trees	0.9104 F1
2nd	Rotation Forest Classifier	Bagging & randomized ensemble	0.9050 F1
3rd	Bagging Classifier	Bagging ensemble	0.9049 F1
4	Random Subspace Classifier	Bagging & randomized ensemble	0.9025 F1
5	Sequence-to-One Transformer	Sequential transformer	0.9021 F1
6	LightGBM Classifier	Gradient boosting	0.9019 F1
7	Balanced Random Forest Classifier	Balanced tree ensemble	0.9015 F1
8	XGBoost Classifier	Gradient boosting	0.8999 F1
9	Informer	Sequential time-series transformer	0.8994 F1
10	Random Forest Classifier	Tabular ensemble / trees	0.8975 F1

Operational meaning. The winning Extra Trees classifier balances high precision and recall, making it suitable for alerting shipments likely to exceed the severe-delay threshold.

Threshold note. The decision threshold should be re-calibrated if the business changes the SLA definition or the cost of false alerts versus missed delays.

Delay Propagation Model Ranking

Top 10 completed results for checkpoint/prefix prediction
Predicting final delay after partial journey observation

Task definition. At each observed checkpoint or prefix, the model predicts **final_delay_minutes**. Tabular propagation models are primarily evaluated by **MeanCheckpointMAE** over 25%, 50%, 75%, and 100% progress checkpoints.

3. Delay Propagation - Top 10 Completed Results

Ranking metric: **MAE**; lower is better. The protocol column is kept because not all completed neural models were scored using the same checkpoint protocol.

Rank	Model	Metric / protocol	Best score
1st	Extra Trees Regressor	v6 tabular checkpoint protocol	95.484 MAE
2nd	LightGBM Regressor	v6 tabular checkpoint protocol	111.345 MAE
3rd	HistGradientBoosting Regressor	v6 tabular checkpoint protocol	111.905 MAE
4	Random Forest Regressor	v6 tabular checkpoint protocol	114.264 MAE
5	XGBoost Regressor	v6 tabular checkpoint protocol	115.600 MAE
6	CatBoost Regressor	v6 tabular checkpoint protocol	123.096 MAE
7	Decision Tree Regressor	v6 tabular checkpoint protocol	124.554 MAE
8	Gradient Boosting Regressor	v6 tabular checkpoint protocol	128.607 MAE
9	Sequence-to-Sequence Transformer	Sequential prefix-sample protocol	132.101 MAE
10	Temporal Convolutional Network (TCN)	Sequential prefix-sample protocol	140.382 MAE

Strongest comparable result. Extra Trees has the lowest MeanCheckpointMAE among models evaluated under the same v6 checkpoint protocol.

Scientific caveat. Ranks 9-10 are sequential prefix-MAE results. They are useful references, but should not be interpreted as perfectly comparable with MeanCheckpointMAE.

- Checkpoint-aware features
- Trip-balanced weighting
- Progress levels: 25/50/75/100%
- Completed outputs only

Recommended wording for the report: "Extra Trees Regressor achieved the best delay-propagation result among completed models under the homogeneous v6 MeanCheckpointMAE protocol."